

The Sobolev Style Metric:

$$\mathcal{H}^1 = \mathcal{L}^2 + \gamma \mathcal{L}^{2,1} \quad \gamma \propto \frac{|\mathcal{C}|}{|\mathcal{M}|}$$

$$E_{\mathcal{C}}(\mathbf{x}, \mathbf{n}) = \int_{\mathbf{p} \in \mathcal{C}} (\|(\mathbf{p} - \mathbf{x})^\top \mathbf{n}\|^2 + \gamma \|\mathbf{n}(\mathbf{p}) - \mathbf{n}\|^2) d\mathbf{p}$$

$$E_{\mathcal{C}}(\mathbf{x}, \mathbf{n}) = \int_{\mathbf{p} \in \mathcal{C}} (\mathbf{n}^\top (\mathbf{p} - \mathbf{x})(\mathbf{p} - \mathbf{x})^\top \mathbf{n}) d\mathbf{p} + \gamma \int_{\mathbf{p} \in \mathcal{C}} (\mathbf{n}(\mathbf{p}) - \mathbf{n})^\top (\mathbf{n}(\mathbf{p}) - \mathbf{n}) d\mathbf{p}$$

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} E_{\mathcal{C}}(\mathbf{x}) = \arg \min_{\mathbf{x}} \int_{\mathbf{p} \in \mathcal{C}} (\mathbf{p} - \mathbf{x})(\mathbf{p} - \mathbf{x})^\top d\mathbf{p} = \bar{\mathbf{x}}$$

$$E_{\mathcal{C}}(\mathbf{n}) = \mathbf{n}^\top \mathbf{M}(\mathcal{C}) \mathbf{n} + \gamma \int_{\mathbf{p} \in \mathcal{C}} (\mathbf{n}(\mathbf{p})^\top \mathbf{n}(\mathbf{p}) - \mathbf{n}(\mathbf{p})^\top \mathbf{n} - \mathbf{n}^\top \mathbf{n}(\mathbf{p}) + \mathbf{n}^\top \mathbf{n}) d\mathbf{p}$$

$$\mathbf{n}^\top \mathbf{n} = 1, \quad \mathbf{n}(\mathbf{p})^\top \mathbf{n} = \mathbf{n}^\top \mathbf{n}(\mathbf{p})$$

$$E_{\mathcal{C}}(\mathbf{n}) = \mathbf{n}^\top \mathbf{M}(\mathcal{C}) \mathbf{n} + \gamma \int_{\mathbf{p} \in \mathcal{C}} (\mathbf{n}(\mathbf{p})^\top \mathbf{n}(\mathbf{p}) - 2\mathbf{n}(\mathbf{p})^\top \mathbf{n} + 1) d\mathbf{p}$$

$$\mathbf{m} = \int_{\mathbf{p} \in \mathcal{C}} \mathbf{n}(\mathbf{p}) d\mathbf{p}$$

$$E_{\mathcal{C}}(\mathbf{n}) = \mathbf{n}^\top \mathbf{M}(\mathcal{C}) \mathbf{n} - 2\gamma \mathbf{m}^\top \mathbf{n} + (\gamma \|\mathbf{m}\|^2 + \gamma |\mathcal{C}|)$$

$$\mathbf{n}^* = \arg \min_{\mathbf{n}} E_{\mathcal{C}}(\mathbf{n}) = \arg \min_{\mathbf{n}} (\mathbf{n}^\top \mathbf{M}(\mathcal{C}) \mathbf{n} - 2\gamma \mathbf{m}^\top \mathbf{n})$$

$$\mathcal{L}(\mathbf{n}, \mu) = \mathbf{n}^\top \mathbf{M}(\mathcal{C}) \mathbf{n} - 2\gamma \mathbf{m}^\top \mathbf{n} + \mu(\mathbf{n}^\top \mathbf{n} - 1)$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{n}} = 2\mathbf{M}(\mathcal{C}) \mathbf{n} - 2\gamma \mathbf{m} + 2\mu \mathbf{n} = 2\mathbf{M}(\mathcal{C}) + 2\mu \mathbf{I} - 2\gamma \mathbf{m} = 0$$

$$\Rightarrow (\mathbf{M}(\mathcal{C}) + \mu \mathbf{I}) \mathbf{n} = \gamma \mathbf{m}$$

$$\frac{\partial \mathcal{L}}{\partial \mu} = \mathbf{n}^\top \mathbf{n} - 1 = 0$$

$$\rho = -\mu$$

$$(\mathbf{M}(\mathcal{C}) - \rho \mathbf{I}) \mathbf{n} = \gamma \mathbf{m}$$

$$\mathbf{M}(\mathcal{C}) = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top, \mathbf{I} = \mathbf{U}\mathbf{U}^\top = \mathbf{U}^\top\mathbf{U}$$

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}, \;\; \lambda_1 \leq \lambda_2 \leq \lambda_3$$

$$\boldsymbol{m} = \mathbf{U}\boldsymbol{a}$$

$$\boldsymbol{n} = \mathbf{U}\boldsymbol{b}$$

$$(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top - \rho \mathbf{I})\mathbf{U}\boldsymbol{b} = \gamma \mathbf{U}\boldsymbol{a}$$

$$\Rightarrow \mathbf{U}\mathbf{\Lambda}\boldsymbol{b} - \rho \mathbf{U}\boldsymbol{b} = \gamma \mathbf{U}\boldsymbol{a}$$

$$\Rightarrow (\mathbf{\Lambda} - \rho \mathbf{I})\boldsymbol{b} = \gamma \boldsymbol{a}$$

$$\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} - \rho \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \gamma \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\begin{bmatrix} \lambda_1 b_1 \\ \lambda_2 b_2 \\ \lambda_3 b_3 \end{bmatrix} - \rho \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \gamma \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\lambda_k b_k - \rho b_k = \gamma a_k$$

$$(\lambda_k - \rho) b_k = \gamma a_k$$

$$b_k = \frac{\gamma a_k}{(\lambda_k - \rho)}$$

$$\boldsymbol{n} = \mathbf{U} \begin{bmatrix} \gamma a_1 (\lambda_1 - \rho)^{-1} \\ \gamma a_2 (\lambda_2 - \rho)^{-1} \\ \gamma a_3 (\lambda_3 - \rho)^{-1} \end{bmatrix}$$

$$f(\rho) = \gamma^2 \sum_{k=1}^3 (a_k^2 (\lambda_k - \rho)^{-2}) - 1 = 0$$

$$\lambda_1 \leq \lambda_2 \leq \lambda_3$$

$$\lim_{\rho \rightarrow -\infty} f(\rho) = -1 < 0 \, , \, \lim_{\rho \rightarrow \lambda_1^-} f(\rho) = \infty > 0$$